Activity: measuring AI speedup

Group members:

AI speedup?

To investigate the effect of AI on user productivity, researchers conducted a randomized experiment with 16 experienced software developers. The authors specified 246 different software development tasks, and randomly assigned each task to be completed with or without AI. The tasks were then divided between the different developers, and the researchers recorded the time it took to complete each task.

Each row in the data represents one task. We will work with the following variables:

- `dev_id`: unique ID for each developer
- `predicted_time_no_ai`: How long the developer estimated the issue would take *without* AI (minutes)
- `ai_treatment`: 0 if AI use was not allowed, 1 if AI use was allowed
- `initial_implementation_time`: Time taken by the developer to solve the issue and create an initial pull request (an initial solution) (minutes)

Models and diagnostics

The following pages show two models, their output, and accompanying diagnostics. For all the diagnostics, "residuals" is the standardized least squares conditional residuals. (Using least squares conditional residuals because of confounding with the random effects, and standardizing so we can check for constant variance).

Model 1

$$\mathrm{T}_{ij} = (\beta_0 + a_i) + (\beta_1 + b_i)\mathrm{AI}_{ij} + \beta_2 \mathrm{EstTimeNoAI}_{ij} + \varepsilon_{ij}$$

$$\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2) \qquad \begin{bmatrix} a_i \\ b_i \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_a^2 & \rho_{ab}\sigma_a\sigma_b \\ \rho_{ab}\sigma_a\sigma_b & \sigma_b^2 \end{bmatrix} \right)$$

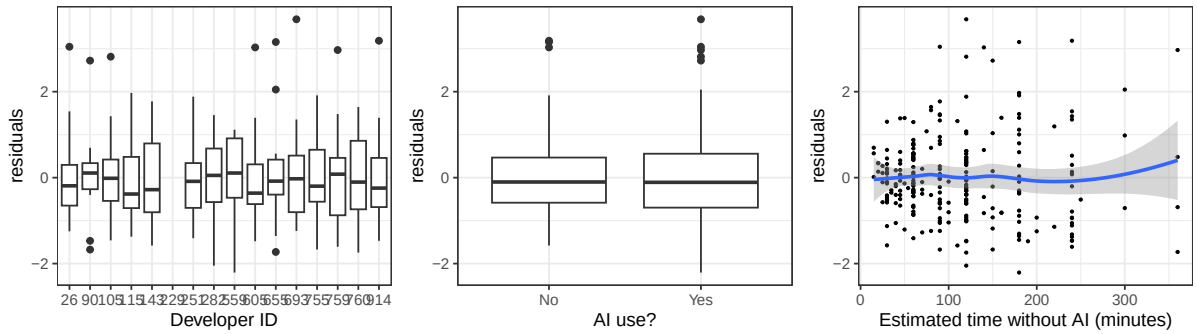
where T_{ij} is the initial completion time for developer i on task j, AI_{ij} is 1 if AI use was allowed on the task, and $\mathrm{EstTimeNoAI}_{ij}$ is the developer's guess for the time it would take to complete the task without AI (the guesses were made before being assigned the tasks).

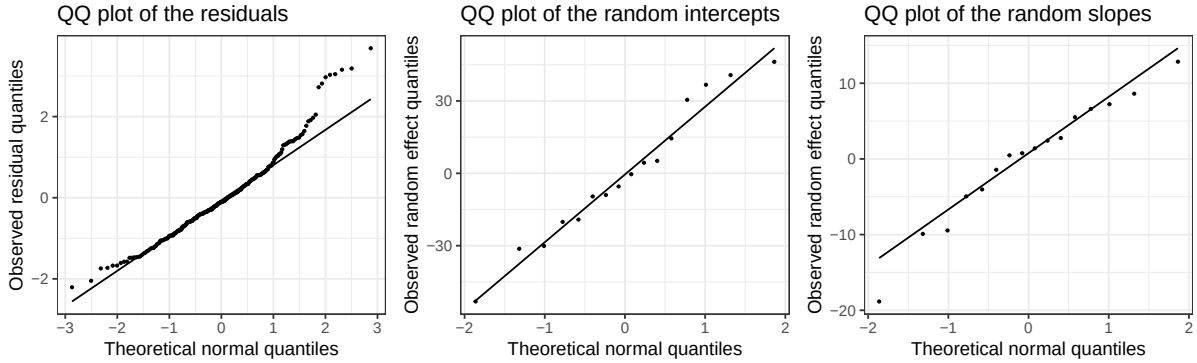
```
m1 <- lmer(initial_implementation_time ~
             ai_treatment + predicted_time_no_ai +
             (ai_treatment|dev_id), data = metr_ai)

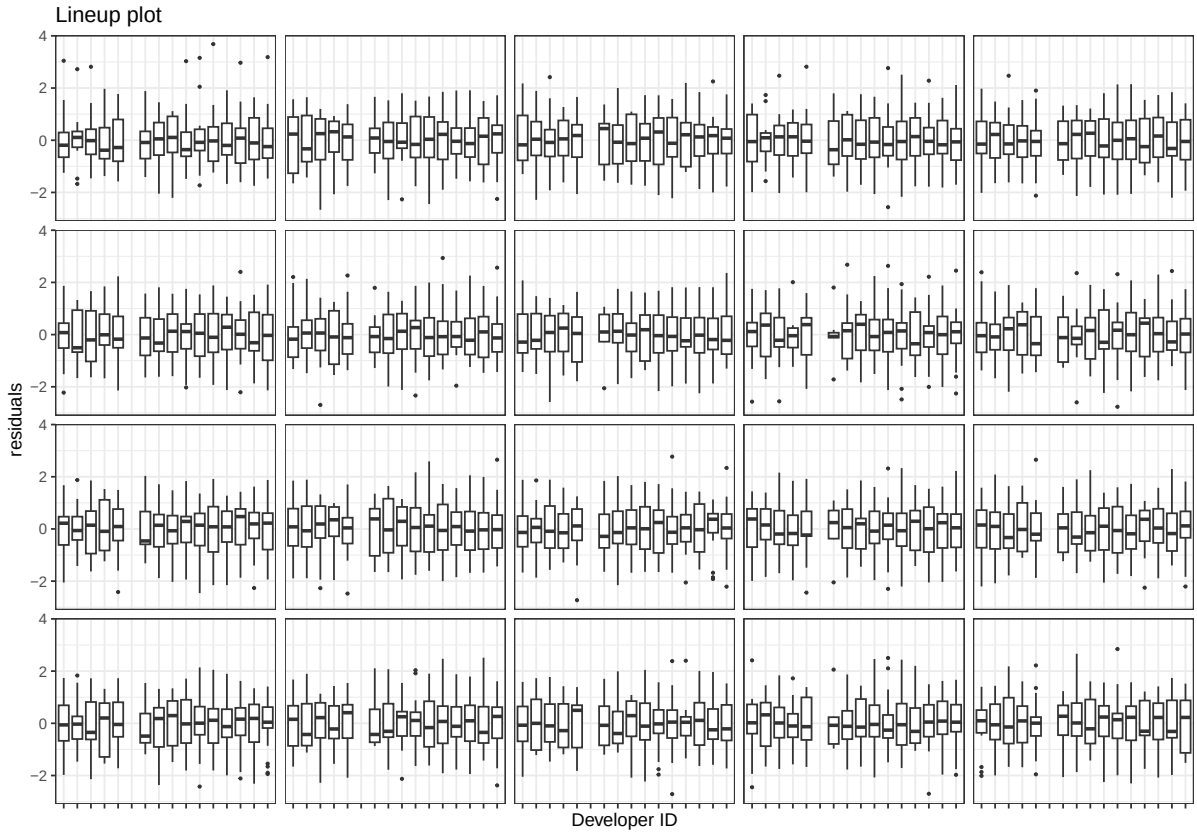
summary(m1)
```

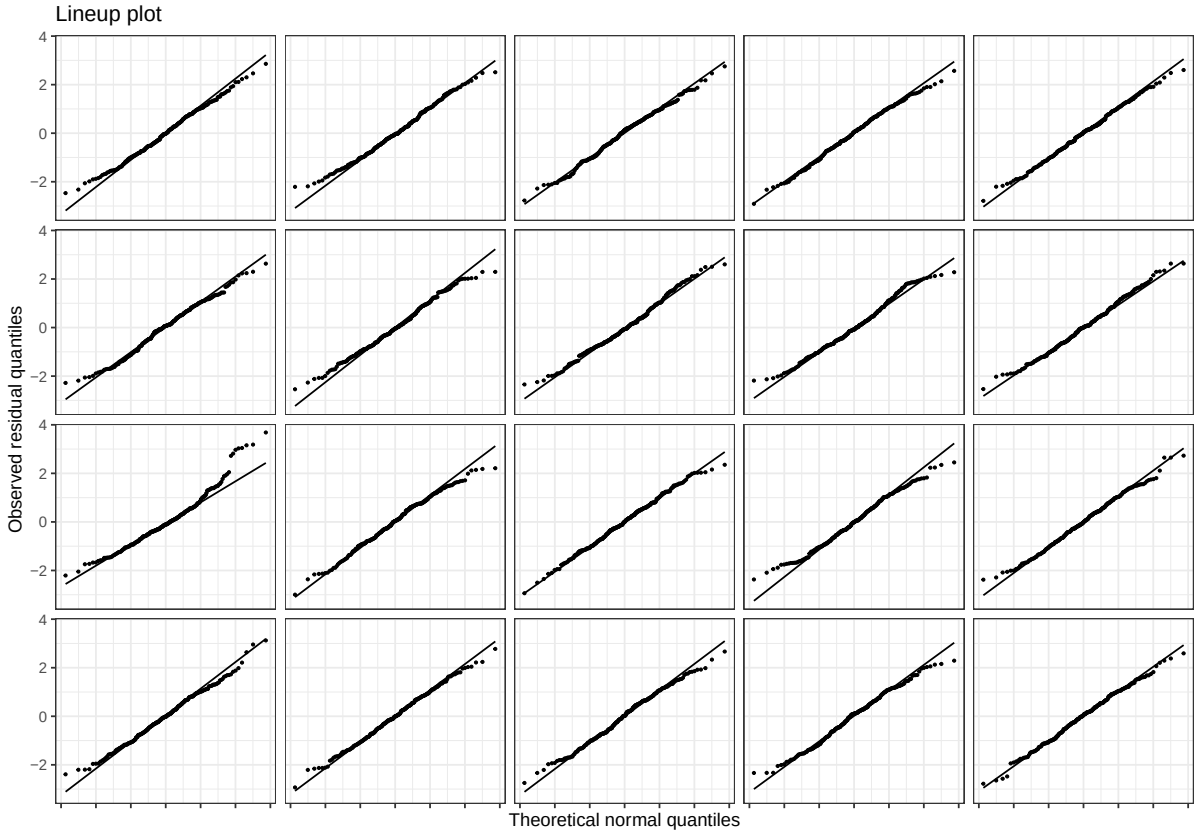
```
Random effects:
 Groups   Name         Variance Std.Dev. Corr
 dev_id   (Intercept)  1205.2   34.72
          ai_treatment  352.9   18.79    -0.19
 Residual              5692.8   75.45
Number of obs: 246, groups:  dev_id, 16

Fixed effects:
                     Estimate Std. Error t value
(Intercept)          -18.57734   13.96529  -1.330
ai_treatment          17.36730   11.13830   1.559
predicted_time_no_ai   1.05437    0.07498  14.062
```


residuals
Developer ID
AI use?
No
Yes
Estimated time without AI (minutes)


QQ plot of the residuals
Observed residual quantiles
Theoretical normal quantiles
QQ plot of the random intercepts
Observed random effect quantiles
Theoretical normal quantiles
QQ plot of the random slopes
Observed random effect quantiles
Theoretical normal quantiles


Lineup plot
residuals
Developer ID


Lineup plot
Observed residual quantiles
Theoretical normal quantiles
4
2
0
−2

Model 2

$$\log(\mathrm{T}_{ij}) = (\beta_0 + a_i) + (\beta_1 + b_i)\mathrm{AI}_{ij} + \beta_2 \log(\mathrm{EstTimeNoAI}_{ij}) + \varepsilon_{ij}$$

$$\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2) \qquad \begin{bmatrix} a_i \\ b_i \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_a^2 & \rho_{ab}\sigma_a\sigma_b \\ \rho_{ab}\sigma_a\sigma_b & \sigma_b^2 \end{bmatrix} \right)$$

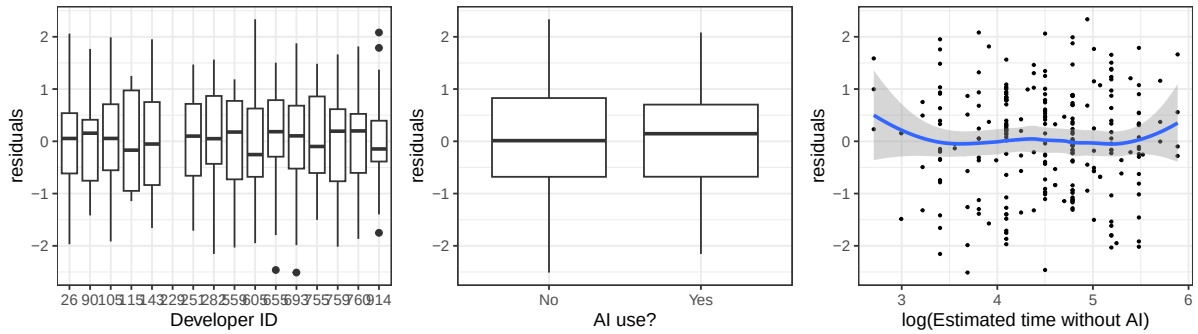
where T_{ij} is the initial completion time for developer i on task j, AI_{ij} is 1 if AI use was allowed on the task, and $\mathrm{EstTimeNoAI}_{ij}$ is the developer's guess for the time it would take to complete the task without AI (the guesses were made before being assigned the tasks).

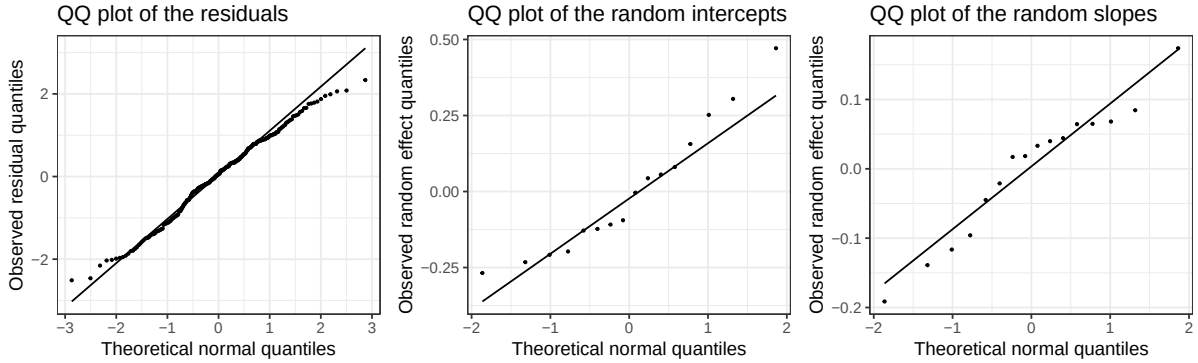
```
m2 <- lmer(log(initial_implementation_time) ~
             ai_treatment + log(predicted_time_no_ai) +
             (ai_treatment|dev_id), data = metr_ai)

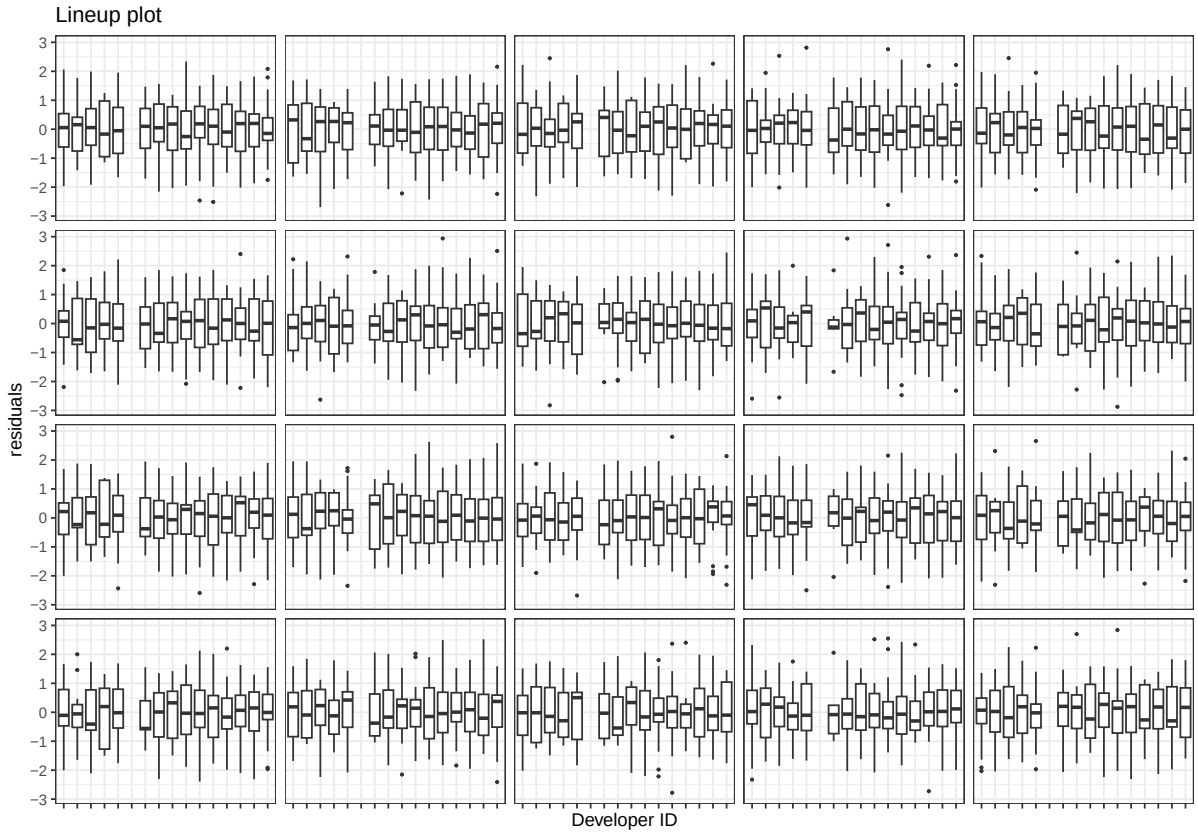
summary(m2)
```

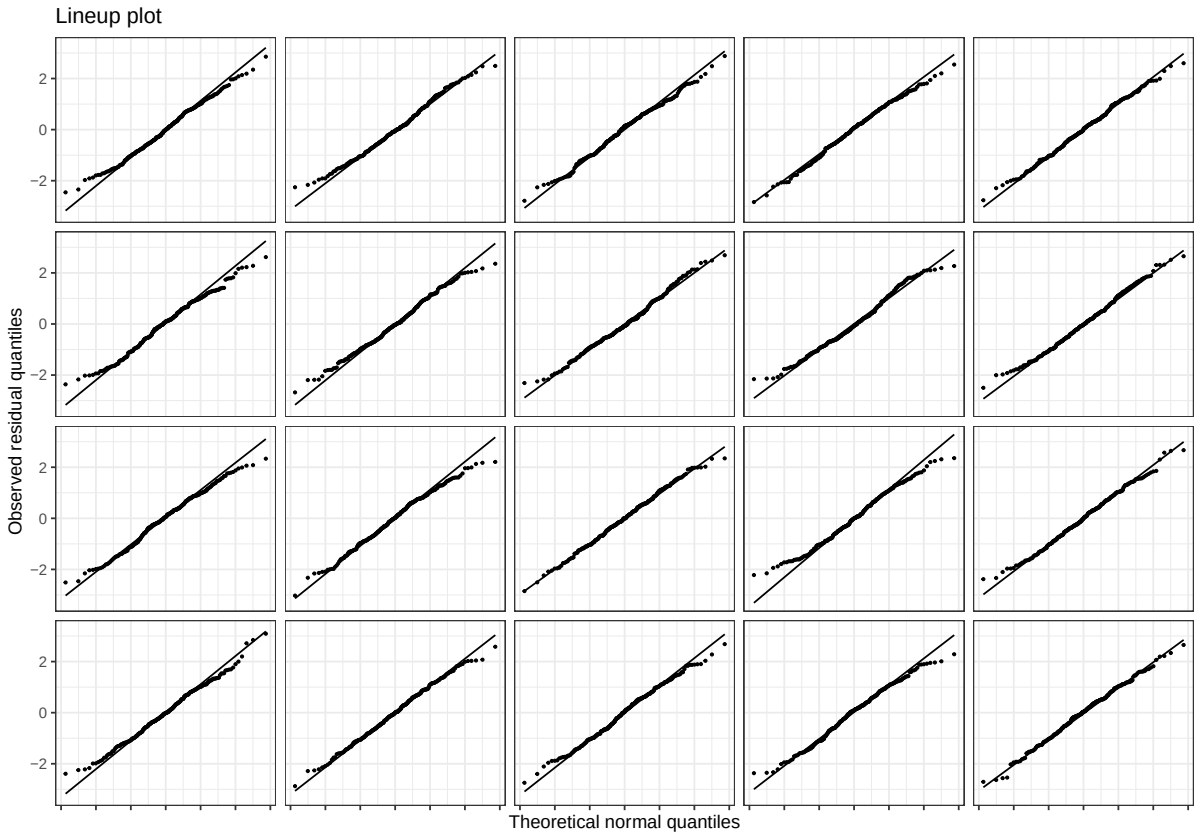
```
Random effects:
 Groups   Name         Variance Std.Dev. Corr
 dev_id   (Intercept)  0.07439  0.2727
          ai_treatment 0.03426  0.1851   -0.57
 Residual              0.35623  0.5969
Number of obs: 246, groups:  dev_id, 16

Fixed effects:
                    Estimate Std. Error t value
(Intercept)          -0.17707    0.28860  -0.614
ai_treatment          0.15577    0.09261   1.682
log(predicted_no_ai)  0.97549    0.06155  15.849
```


residuals
Developer ID
AI use?
No
Yes
log(Estimated time without AI)


QQ plot of the residuals
Observed residual quantiles
Theoretical normal quantiles
QQ plot of the random intercepts
Observed random effect quantiles
Theoretical normal quantiles
QQ plot of the random slopes
Observed random effect quantiles
Theoretical normal quantiles


Lineup plot
residuals
Developer ID


Lineup plot
Observed residual quantiles
Theoretical normal quantiles
2
0
−2